ModelIt!

Teacher's Guide

How Do Pandemics Spread?

Modeling Epidemiological Dynamics, Transmission Networks, and Intervention Strategies

Quick Reference

Grade Level	Grade 12L3
NGSS Standard	HS-LS2-8
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS2-8

Evaluate the evidence for the role of group behavior on individual and species' chances to survive and reproduce.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use mathematical models of exponential growth, reproduction numbers, and herd immunity thresholds to predict pandemic trajectories and evaluate intervention effectiveness quantitatively.
Disciplinary Core Idea	**LS2.D Social Interactions and Group Behavior** — Students model how individual behaviors — contact patterns, quarantine compliance, vaccination decisions — aggregate to determine population-level pandemic outcomes, demonstrating that group behavior emerges from but cannot be predicted by individual actions alone.
Crosscutting Concept	**Stability and Change** — Students analyze how pandemic systems can shift rapidly from apparent stability (low case counts) to explosive growth (exponential phase) and identify the tipping points where interventions can restore stability or where delay makes control impossible.

CAST Testing Alignment

MULTIPLE CHOICE:

A new virus with R0 = 4 is spreading in a city. Social distancing reduces contacts by 40% and masks reduce transmission probability by 30%. If both are implemented, what is the approximate effective reproduction number, and is this sufficient to stop exponential growth?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain why a city that delays implementing social distancing by just three weeks during the early phase of a pandemic might experience five times more total deaths than a city that acts immediately. Reference the mathematics of exponential growth and at least two model components.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Pandemics have shaped human history more profoundly than wars or political revolutions. The 1918 influenza killed 50-100 million people. HIV/AIDS has killed over 40 million since 1981. COVID-19 caused over 20 million excess deaths worldwide. Yet despite this devastating history, most people struggle to understand why pandemics are so dangerous — because the human brain is not wired to intuitively grasp exponential growth.

The Mathematics of Exponential Growth

If one person infects three others (R0 = 3) and each of those three infects three more, the chain grows from 1 to 3 to 9 to 27 to 81 to 243 to 729 in just six generations. After 20 generations — which can occur in as few as 40-60 days for a respiratory virus — the theoretical number exceeds 3.5 billion. In reality, the growth slows as the susceptible population shrinks, but the key insight is that by the time exponential growth becomes visible (hundreds or thousands of cases), millions of transmission chains are already in progress. This is why epidemiologists sound the alarm when cases number in the dozens — they can see the mathematical future that is invisible to intuition.

The SIR Model Framework

Epidemiologists use compartmental models to track pandemic dynamics. The simplest is SIR: Susceptible, Infected, Recovered. Everyone begins as Susceptible. Infected individuals transmit at rate R0 and eventually move to Recovered (immune) or die. The pandemic ends when enough people have moved from S to R that the remaining susceptible population is too small to sustain transmission — this is herd immunity. The herd immunity threshold is 1 - 1/R0: for R0 = 3, it is 67%; for R0 = 5, it is 80%. More sophisticated models add Exposed (pre-infectious) and Deceased compartments, age structure, spatial heterogeneity, and waning immunity.

Intervention Mechanics

Social distancing reduces the contact rate component of R0. Masks reduce the transmission probability per contact. Quarantine removes infectious individuals from the transmission chain. Vaccination moves people directly from Susceptible to Recovered without experiencing disease. Each intervention has a quantifiable effect on the effective reproduction number (Reff). The goal is to push Reff below 1.0, at which point each infected person transmits to fewer than one other on average and the epidemic declines. The challenge is that interventions have social, economic, and psychological costs that limit duration and compliance.

Equity and Pandemic Impact

Pandemics do not affect all communities equally. COVID-19 mortality rates were 2-3 times higher in Black, Latino, and Indigenous communities in the United States due to structural factors: higher rates of essential worker employment, crowded housing that prevents effective isolation, less access to healthcare, and higher prevalence of underlying conditions caused by chronic stress and environmental exposures. Effective pandemic preparedness must explicitly address these disparities — equitable vaccine distribution, economic support for quarantine compliance, and targeted testing in underserved communities are not just ethical imperatives but epidemiological necessities, because uncontrolled transmission in any subpopulation threatens the entire population.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify ten key components of the pandemic system: viral transmissibility, population density, initial immunity, contact rate, infection detection, quarantine compliance, vaccination coverage, healthcare capacity, mutation rate, and cumulative mortality.

E **Establish Relationships**

Students map the causal relationships showing how R0 drives exponential growth, how interventions reduce the effective reproduction number, and how healthcare capacity creates a mortality threshold effect.

V **Visualize & Model**

Students build a ten-component computational model showing the competing dynamics between viral transmission and intervention effectiveness, with feedback loops from mutation and waning immunity.

E **Evaluate Outcomes**

Students run uncontrolled spread, early intervention, and vaccine rollout scenarios to quantify how timing and intensity of interventions determine total mortality.

R **Revise & Extend**

Students add economic impact, misinformation spread, or genomic surveillance to model how social and informational factors modify the biological dynamics of pandemic spread.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"In January 2020, a handful of mysterious pneumonia cases appeared in Wuhan, China. Twelve months later, COVID-19 had infected over 100 million people and killed over 2 million. How does a single case become a global catastrophe? The answer is mathematics — and today you will learn to read the equations that predict pandemics."*

Watch For

Show cover image. Ask: how many of you were personally affected by COVID-19? Acknowledge the lived experience in the room before moving to the science.

Slide 2: Learning Objectives

3 min

Say: *"Today you are building a mathematical model of pandemic spread. You will discover why exponential growth is so deceptive, why timing matters more than intensity, and why equity is an epidemiological variable, not just a moral one."*

Watch For

Have students read objectives. Pre-teach R0 with a physical demonstration: give one student 3 cards, each of those students gets 3 cards, and count how fast the classroom is 'infected.'

Slide 3: Big Question

3 min

Say: *"Small differences in transmission rate or intervention timing can mean the difference between thousands and millions of deaths. Why? Because your brain cannot intuitively understand exponential growth. Today your model will teach your brain what it cannot feel."*

Watch For

Quick-write: Students estimate how many people one infected person with R0=3 would ultimately infect after 15 transmission generations. Reveal the answer (over 14 million) to demonstrate exponential growth intuition failure.

Slide 4: LEVER Framework

2 min

Say: *"We will Label ten components of the pandemic system, Establish their relationships, Visualize how they interact, Evaluate intervention scenarios, and Refine by adding equity and information variables."*

Watch For

Preview each LEVER step. Emphasize that this model will explain why the same virus can kill 1,000 in one city and 100,000 in another.

Slide 5: Activity 1: Components

8 min

Say: *"Three of these components describe the virus and the population it enters — the inputs we cannot control initially. Seven describe the human response — the variables we CAN control. Sorting them correctly is the first step to understanding why intervention design matters."*

Watch For

Students sort external versus internal components. Debate: is Contact Rate external or internal? It reflects both cultural norms and policy choices. Use this to discuss the boundary between behavior and biology.

Slide 6: Activity 2: Connections

8 min

Say: *"R0 drives infection rate. Contact rate multiplies transmission. Quarantine removes infectious people from circulation. Vaccination shrinks the susceptible pool. But these all interact — and some relationships create reinforcing feedback loops that can make a pandemic spiral out of control."*

Watch For

Students draw arrows showing positive and negative relationships. Identify the critical feedback loop: overwhelmed healthcare -> higher mortality -> public fear -> better compliance but also economic collapse -> reduced compliance.

Slide 7: Activity 3: Simulation

10 min

Say: *"Now run three scenarios: uncontrolled spread, early intervention, and vaccine rollout. Record your predictions BEFORE you run each scenario. The gap between what you predict and what happens will teach you something important about your intuition."*

Watch For

Students run all scenarios, recording predictions versus observations. Key question after each: how close was your prediction? Most students will underestimate exponential growth and underestimate the impact of early intervention.

Slide 8: Discoveries

5 min

Say: *"Your model reveals a brutal truth: in a pandemic, every day of delay costs lives — not linearly, but exponentially. The difference between acting at 100 cases and acting at 1,000 cases is not 10x more deaths. It can be 50x or 100x more deaths."*

Watch For

Class discussion of discoveries. Show the comparative mortality graphs from the three scenarios. Ask: knowing this, why do governments often delay action?

Slide 9: STEM Challenge

12 min

Say: *"A novel virus with R0 of 4 is approaching your city. You have 2-4 weeks before the first case arrives. No vaccine exists. Design a preparedness plan that saves the most lives while keeping the city functioning. Your model data is your evidence."*

Watch For

Groups design preparedness plans with specific timelines, trigger points, and interventions. Each team must justify every decision with model evidence. Present and compare: which plan saves the most lives? Which is most equitable?

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Viral Transmissibility (R0), Population Density, Initial Immune Fraction
- **INTERNAL:** Contact Rate, Infection Detection Rate, Quarantine Compliance, Vaccination Coverage Rate, Healthcare System Capacity, Mutation Rate, Cumulative Mortality

Relationships

- **Viral Transmissibility to Cumulative Mortality:** POSITIVE (+) — Higher R0 means faster exponential growth, more infections before interventions take effect, and greater likelihood of overwhelming healthcare capacity, all of which increase total deaths.
- **Population Density to Contact Rate:** POSITIVE (+) — Denser populations have more frequent close contacts in housing, transportation, workplaces, and public spaces, directly increasing the number of transmission opportunities per infected person per day.
- **Contact Rate to Cumulative Mortality:** POSITIVE (+) — Higher contact rates amplify the effective reproduction number, accelerating transmission and increasing the total number of infections. Reducing contact rate through social distancing is one of the most powerful intervention levers.
- **Infection Detection Rate to Quarantine Compliance:** POSITIVE (+) — Faster detection of infected individuals enables earlier quarantine of cases and contacts. Without detection, quarantine is impossible; with rapid detection, transmission chains can be broken before secondary spread occurs.
- **Vaccination Coverage Rate to Cumulative Mortality:** NEGATIVE (-) — Higher vaccination coverage reduces the susceptible population, lowering the effective reproduction number. Once vaccination exceeds the herd immunity threshold, sustained transmission becomes impossible and mortality stabilizes.
- **Healthcare System Capacity to Cumulative Mortality:** NEGATIVE (-) — Greater healthcare capacity means more patients receive treatment, reducing the case fatality rate. When capacity is exceeded, treatable patients die from lack of care, and mortality rates can increase 3-5 fold.
- **Mutation Rate to Vaccination Coverage Rate:** NEGATIVE (-) — Higher mutation rates increase the probability of immune-evasive variants that reduce vaccine effectiveness, potentially requiring booster campaigns and resetting progress toward herd immunity.
- **Quarantine Compliance to Cumulative Mortality:** NEGATIVE (-) — Higher quarantine compliance removes infectious individuals from the transmission chain faster, reducing secondary infections. Even moderate improvements in compliance have large effects because they break exponential transmission chains.

Key Discoveries

- Exponential growth is deceptively slow at first and then explosively fast — by the time a pandemic is visibly widespread, the window for effective early intervention has already closed, which is why mathematical models are essential for triggering preemptive action
- The timing of interventions matters more than their intensity — implementing moderate social distancing two weeks after the first case prevents more deaths than implementing strict lockdowns six weeks later, because exponential growth makes every day of delay dramatically more costly
- Healthcare capacity creates a mortality multiplier — when hospitals are overwhelmed, deaths increase not only from the pandemic disease but from all other conditions that cannot receive treatment, meaning the true death toll far exceeds direct infection mortality

- Herd immunity is not a fixed threshold but a moving target — viral mutations that increase transmissibility raise the herd immunity threshold, and variants that evade immunity reset the susceptible population, meaning a pandemic can have multiple waves even with high vaccination rates

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Herd immunity will happen naturally so we don't need vaccines"	Natural herd immunity requires a large fraction of the population to become infected, which means millions of severe cases and hundreds of thousands of deaths for a disease with even modest fatality rates. For a disease with R0=3 and 1% fatality rate in a country of 330 million, natural herd immunity would require approximately 220 million infections and 2.2 million deaths. Vaccination achieves the same immunological endpoint without the mass casualties.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Pandemics end when the virus runs out of people to infect"	This is partially true but oversimplified. Pandemics can have multiple waves because immunity wanes over time, new susceptible individuals are born, and viral mutations create variants that evade existing immunity. COVID-19 demonstrated this with successive waves driven by Alpha, Delta, and Omicron variants, each partially escaping prior immunity. A pandemic transitions to endemic status when population immunity stabilizes transmission at a sustainable baseline level.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Masks and social distancing don't work because the pandemic continued even with those measures"	Individual interventions reduce but do not eliminate transmission. The goal is not zero transmission but reducing the effective reproduction number below 1. If R0 is 4 and social distancing reduces contacts by 40% and masks reduce per-contact transmission by 30%, the combined effect reduces Reff to approximately 1.7 — still above 1, meaning the pandemic continues but at a much slower rate with far fewer total deaths. The fact that cases continued does not mean interventions failed; it means the virus was so transmissible that multiple simultaneous interventions were needed.